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Clinical features and prognosis of Miller Fisher syndrome

Article abstract—The authors reviewed the clinical features and outcome of Miller Fisher syndrome (MFS) for 50 consecutive patients with MFS including 28 patients who received no immunotherapy. Besides the characteristic clinical triad (ophthalmoplegia, ataxia, and areflexia), pupillary abnormalities, blepharoptosis, and facial palsy are frequent in MFS, whereas sensory loss is unusual despite the presence of profound ataxia. Patients with MFS usually had good recovery and no residual deficits.

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Masahiro Mori, MD; Satoshi Kuwabara, MD; Toshio Fukutake, MD; Nobuhiro Yuki, MD, PhD; and Takamichi Hattori, MD

Miller Fisher syndrome (MFS) is characterized by the clinical triad of ophthalmoplegia, ataxia, and areflexia, and is considered a variant form of Guillain–Barré syndrome (GBS).[1] Because MFS is a rare disorder, and few studies have been made of a large number of consecutive patients with MFS,[2] the frequency and characteristics of neurologic symptoms and signs and natural history and prognosis are not fully understood. To shed light on these clinical issues, we reviewed data for 50 consecutive patients with MFS.

Patients and methods. *Patients.* Fifty-three patients with MFS and 148 patients with GBS were seen at our hospitals between 1979 and 1999. All 53 patients with MFS with the clinical triad of MFS (ophthalmoplegia, ataxia, and areflexia) and acute onset without major limb weakness or other signs suggestive of CNS involvement were selected. Three of the 53 patients later developed profound limb weakness and were excluded from analysis.

Assessments of clinical recovery. Twenty-two of the 50 patients with MFS received plasmapheresis, and 28 patients received no immunotherapy. We followed the clinical courses of the latter 28 patients focusing on amelioration of the clinical triad, except areflexia. We also assessed the period to the beginning of recovery and for the disappearance of ophthalmoplegia and ataxia from time of onset. Ataxia severity was evaluated on the Hughes functional grading scale.[3]

Results. *Clinical features.* The median age at the time of MFS onset was 40 years (range 13 to 78 years) for the 34 men (68%) and 16 women (32%). MFS clearly was seasonal, predominating in spring (March to May).

Of the 50 patients, respiratory symptoms occurred within 1 month before MFS onset in 76%, gastrointestinal symptoms in 4%, fever in 2%, and no infectious symptoms in 18%. The median duration of infectious symptoms for 21 patients was 7 days (range 1 to 26 days), and the median interval between the onset of infection and neurologic symptoms was 8 days (range 1 to 30 days).

MFS frequently started with diplopia (78%) or ataxia (46%) or both (34%) on the same day. Other initial symptoms were dysesthesia in the limbs (14%), dysphagia (2%), blepharoptosis (2%), and photophobia (2%). The median interval between the onset of diplopia and ataxia was 1 day (range 0 to 4 days).

The median time to the nadir of symptoms after the neurologic onset of MFS was 6 days (range 2 to 21 days). The neurologic symptoms are given in the table, at which time there was complete external ophthalmoplegia in 15 patients (30%), and 15 patients (30%) could not walk independently owing to ataxia. The ataxia grade was 4 in 7 (14%) patients, 3 in 8 (16%), 2 in 28 (56%), and 1 in 7 (14%).

Of the 27 patients whose ophthalmoplegia was examined monthly, few had findings suggestive of supranuclear involvement: preservation of Bell's phenomenon despite paralysis of voluntary upward gaze in two, gaze-evoked horizontal dissociated nystagmus in two, preservation of convergence despite adduction palsy with conjugate gaze in one, and internuclear ophthalmoplegia in one.

From the Department of Neurology (Drs. Mori, Kuwabara, Fukutake, and Hattori), Chiba University School of Medicine, Chiba, Japan; and Department of Neurology (Dr. Yuki), Dokkyo University School of Medicine, Tochigi, Japan.

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Address correspondence and reprint requests to Dr. Masahiro Mori, Department of Neurology, Chiba University School of Medicine, 1-8-1 Inohana, Chuo-ku, Chiba, 260-8670 Japan; e-mail morim@olive.ocn.ne.jp

Table *Neurologic symptoms and signs at the peak of Miller Fisher syndrome (n = 50)*

Category	Abnormalities	No. (%) of cases
Cranial nerve	Pupillary	21 (42)
	Blepharoptosis	29 (58)
	Facial palsy	16 (32)
	Bulbar palsy	13 (26)
Sensory	Dysesthesia	12 (24)
	Decrease in superficial sensation	10 (20)
	Decrease in vibratory or deep sensation	9 (18)
Motor	Weakness (grade 4 on the MRC scale)	10 (20)
Autonomic	Micturition disturbances	8 (16)

All patients had ophthalmoplegia, ataxia, and areflexia.

MRC = Medical Research Council.

Mydriasis was present in 21 (42%) patients, 10 of whom had anisocoria. Light reflexes were sluggish in 21 (42%) of 45 patients, one of whom had a unilateral abnormality. Peripheral-type facial palsy was found in 16 (32%) patients, in 6 (38%) of whom it had worsened when the other neurologic symptoms were lessening. In one of the six patients, facial palsy developed after the start of plasmapheresis.

Sensory loss, which occurred in 12 (24%) patients, included decreases in superficial sensations in 10 (20%) and decreases in deep sensations in 9 (18%). In contrast, results of the thumb localizing test, a marker for impaired proprioception in the fixed limb,[4] were abnormal in 76% of the 25 patients examined.

We tested the sera of 36 of the 53 patients with MFS; 32 (89%) had elevated anti-GQ1b IgG antibody.

Natural course and prognosis. The natural recovery course in 28 patients with MFS who had received no immunotherapy was followed for a median period of 4 (range 1 to 185) months.

Almost all the patients showed a decrease in ataxia and ophthalmoplegia, and areflexia started to improve, in that order. The respective median (range) periods between neurologic onset and the beginning of recovery of ataxia and ophthalmoplegia were 12 (3 to 41) and 15 (3 to 46) days, and the respective median (range) periods for the disappearance of ataxia and ophthalmoplegia were 32 (8 to 271) and 88 (29 to 165) days. About two-thirds of the patients had areflexia or hyporeflexia by the end of the follow-up, indicative that tendon reflex recovery was much slower than recovery from other deficits.

The chance of recovery from ataxia and ophthalmoplegia based on the Kaplan–Meyer curve was not affected by sex, age, whether there had been prior infection, disability at the peak of illness, whether ophthalmoplegia was complete or incomplete at the peak, or latency to the peak (whether shorter than 1 week).

Six months after neurologic onset, all patients were almost free of ataxia and ophthalmoplegia and had returned to their normal activities.

Discussion. The incidence of MFS is reported to be approximately 1 to 5% that of GBS[5] in Western countries. In contrast, the incidence of MFS in Taiwan is reported to be 19% that of GBS,[6] and our study showed it to be 25% that of GBS in Japan. Considering that the incidence of GBS is about the same all over the world,[5] the incidence of MFS in Asian countries therefore appears to be higher than in Western countries.

Several lines of evidence support molecular mimicry between ganglioside GQ1b and a *Campylobacter jejuni* lipopolysaccharide.[7] Our findings, however, suggest that the majority of MFS cases are associated with a pathogen that causes upper respiratory tract infection, the peak incidence being in spring.

Our results showed pupillary abnormalities, blepharoptosis, and facial palsy are frequent in MFS. Patients with MFS were reported in whom facial palsy developed during plasmapheresis treatment when other neurologic symptoms were lessening.[8] It was suggested that unlike its effect on other neurologic symptoms, plasmapheresis is ineffective for facial palsy. Such a delayed worsening of facial palsy was often seen in this study.

Surprisingly, loss of deep sensation was much less frequent than expected, despite the presence of profound ataxia. Our previous study analyzing postural body sway of 10 patients with MFS showed that most had no sensory loss, but there was evidence of impaired proprioceptive function.[9] Muscle spindle afferents are speculated to be selectively involved in MFS. Our current findings, which confirm that elementary sensory loss is infrequent, are consistent with our prior hypothesis.

The nosologic position of MFS is not clear, with speculation being that the lesion site in MFS is in the peripheral nerves, brainstem, or cerebellum. Our findings showed that the central components (e.g., supranuclear eye movement disorder) might be associated, but only rarely, indicative that the brainstem is not the main lesion site in MFS.

In GBS, older age of the patient, faster disease progression, need for ventilation support, and electrophysiologic evidence of axonal degeneration are correlated with a poor prognosis.[10] We found, however, that sex, age, evidence of a prior infection, disability at the peak of illness, and latency to peak have no effect on the disease's outcome. Irrespective of these factors, ataxia and ophthalmoplegia disappeared a median of 1 and 3 months after onset, and all patients had no or very little disability 6 months after onset. The natural course of MFS is characterized by good recovery.

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Cerebral involvement in graft-versus-host disease after murine bone marrow transplantation

Article abstract—Neurologic manifestation of graft-versus-host disease (GvHD) after allogeneic bone marrow transplantation (BMT) has until now been limited to rare neuromuscular syndromes. Investigating cerebral findings using a murine BMT model, the authors found parenchymal lymphocytic inflammation, microglia activation, and mild cerebral angiitis-like changes in allogeneic transplanted animals but not in syngeneic controls. These findings suggest that cerebral involvement during GvHD may be a new neurologic complication after BMT.

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C.S. Padovan, MD; A. Gerbitz, MD; P. Sostak, MD, PhD; E. Holler, MD; J.L.M. Ferrara, MD; K. Bise, MD; and A. Straube, MD

Allogeneic bone marrow transplantation (BMT) is the treatment of choice for a variety of malignant and nonmalignant conditions. Frequent neurologic complications after BMT include neurotoxicity of immunosuppressant agents, cardioembolic cerebrovascular accidents, metabolic encephalopathy, and CNS infections. Neurologic manifestation of graft-versus-host disease (GvHD), however, which may systemically lead to skin rash, scleroderma-like changes, and liver or bowel disease, has until now been limited to rare neuromuscular syndromes. We recently reported on the occurrence of cerebral angiitis in five allogeneic BMT patients with chronic GvHD.[1] Using a murine BMT model of chronic GvHD, we investigated whether the CNS may be involved in GvHD.

Methods. *BMT.* C57/B6 (H-2^b) mice received either allogeneic BMT from LP (H-2^b) mice (10^6 bone marrow cells and 10^6 splenic cells) or syngeneic BMT from C57/B6 mice after conditioning with total-body irradiation (800 cGy split into two doses), as previously described.[2] The degree of systemic GvHD was assessed weekly in individual animals using a score that incorporates five clinical parameters: weight loss, posture (hunching), activity, fur texture, and skin integrity. Each parameter was graded from 0 to 2; a clinical index was generated by the summation of the five criteria scores (maximum index = 10).[2]

CNS findings were examined 42 days after BMT when a maximum of systemic GVHD score was reached. To differentiate between GvHD and total-body irradiation effects, cerebral findings were also examined at 7 and 14 days.

Immunohistologic staining. After deep anesthesia, the mice were killed, and their brains were fixed in 10% buffered formalin and embedded in paraffin. CNS findings were examined with hematoxylin/eosin-stained slices and immunostained sections using a rat anti-mouse CD45 monoclonal antibody (IgG_{2b}; Pharmingen, San Diego, CA). CD45 antibody recognizes hematopoietic cells (lymphocytes, mononuclear phagocytic cells, no erythrocytes) as well as activated microglia cells. Further immunophenotyping of infiltrating cells, for example, with specific lymphocyte or microglia antibodies, could not be performed because anti-mouse monoclonal antibodies suitable for paraffin-embedded tissue were not available. Immunohistologic staining was performed on 3-μm protease-treated brain sections using a streptavidin/peroxidase method (Dako, Carpinteria, CA). $CD45^+$ cells were examined in the cortex, hippocampus, periventricular area, basal ganglia, pons, and cerebellum by two blinded observers, who

From the Departments of Neurology (Drs. Padovan, Sostak, and Straube) and Neuropathology (Dr. Bise), Ludwig Maximilians University, Munich, and Department of Hematology and Oncology, University of Regensburg (Drs. Gerbitz and Holler), Germany; and Department of Medicine, University of Michigan Medical School and University of Michigan Cancer Center (Dr. Ferrara), Ann Arbor.

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Address correspondence and reprint requests to Dr. C.S. Padovan, Department of Neurology, Ludwig Maximilians University, 81366 Munich, Germany; e-mail: padovan@brain.nefo.med.uni-muenchen.de

